Closing the center-embedding gap in Spanish: can grammatical features do the work of memory?

Norberto Moreno-Quibén[1], Silvia Gumiel-Molina[2], Isabel Pérez-Jiménez[3], and Pilar Pérez-Ocón[4]

[1,2,3,4]Departamento de Filología, Comunicación y Documentación, Universidad de Alcalá (UAH)

Poster Abstract · Cajal Xmas Meeting 2026 · Instituto Cajal, CSIC, Madrid

Abstract

Center embeddings—clauses nested inside one another—push comprehension to its limits, making them a natural probe for the memory architecture that supports language (Karlsson 2007). Their signature is the missing-VP illusion: when one of three required verbs is deleted, readers accept the broken sentence more readily than the intact one, as if a stored prediction had silently decayed (Lewis and Vasishth 2005; Hahn et al. 2022). The illusion was recently demonstrated in Spanish (Pañeda and Lago 2024), with a result mattering beyond linguistics: illusion magnitude tracked the statistical rarity of the verb sequence, while individual working-memory span predicted nothing. Storage capacity, on its own, appears to be the wrong variable.

We ask a question Spanish is uniquely positioned to answer. Spanish offers relativizers that differ in morphological content while occupying identical syntactic positions: featureless *que* versus *el cual* and *quien*, which carry gender, number and animacy. If the illusion arises because degraded memory traces are reconstructed ambiguously, then feature-bearing relativizers should supply distinctive retrieval cues and reduce it—with no change in sentence length, structure, or embedding depth. Featureless *que* should not.

We will test this in 120 speakers, combining speeded acceptability judgements, eye-tracking and EEG, and predict a graded reduction in illusion rate with feature richness, an attenuated sustained anterior negativity during maintenance, and a restored P600 at the offending verb.

The result would place grammatical features, not capacity, at the centre of how the brain holds structure open—and hands a broadly neurobiological question, what makes a memory cue distinctive, a linguistically precise manipulation.

Keywords: sentence processing; cue-based retrieval; working memory; morphosyntactic features; ERP; Spanish

SUPPORTING MATERIALS (NOT PART OF THE ABSTRACT WORD COUNT)

The critical manipulation, on one item. All conditions are double center embeddings of identical depth; the featural content of the relativizer is crossed with grammaticality (VP2 entrevistó present vs. deleted).

(1) a. La política **a la que** el periodista **al que** los reporteros interrumpieron entrevistó dimitió ayer. [GRAMMATICAL]

b. *La política **a la que** el periodista **al que** los reporteros interrumpieron ~~entrevistó~~ dimitió ayer. [UNGRAMMATICAL, MISSING VP2]
BASELINE — *que*, gender/number only via the article; the condition used in all prior work.

(2) a. La política **a la cual** el periodista **al cual** los reporteros interrumpieron entrevistó dimitió ayer. [GRAMMATICAL]

b. *La política **a la cual** el periodista **al cual** los reporteros interrumpieron ~~entrevistó~~ dimitió ayer. [UNGRAMMATICAL, MISSING VP2]
FEATURE-BEARING — *el cual* inflects for gender and number on the pronoun itself.

(3) a. La política **a quien** el periodista **a quien** los reporteros interrumpieron entrevistó dimitió ayer. [GRAMMATICAL]

b. *La política **a quien** el periodista **a quien** los reporteros interrumpieron ~~entrevistó~~ dimitió ayer. [UNGRAMMATICAL, MISSING VP2]
ANIMACY-BEARING — *quien* restricts the antecedent to a human referent.

Differential object marking on *a* blocks a subject-relative reading throughout, so no condition is ambiguous at the relativizer.

REFERENCES

Hahn, Michael et al. (2022). “A resource-rational model of human processing of recursive linguistic structure”. In: *Proceedings of the National Academy of Sciences* 119.43, e2122602119. DOI: 10.1073/pnas.2122602119.

Karlsson, Fred (2007). “Constraints on multiple center-embedding of clauses”. In: *Journal of Linguistics* 43.2, pp. 365–392. DOI: 10.1017/S0022226707004616.

Lewis, Richard L. and Shravan Vasishth (2005). “An activation-based model of sentence processing as skilled memory retrieval”. In: *Cognitive Science* 29.3, pp. 375–419. DOI: 10.1207/s15516709cog0000_25.

Pañeda, Claudia and Sol Lago (2024). “The missing VP illusion in Spanish: Assessing the role of language statistics and working memory”. In: *Open Mind: Discoveries in Cognitive Science* 8, pp. 42–66. DOI: 10.1162/opmi_a_00118.